

Expanding psychometrics with pretrained language models: Evaluating pseudo-factor analysis in applied and multilingual contexts

Simone Varrasi^{a,1}, Giuseppe Alessio Platania^{a,1}, Sabrina Castellano^a, Francesca Ferraioli^{b,*},
Simona Massimino^b, Carmelo Mario Vicario^b, Santo Di Nuovo^{a,2}, Epifanio Damiano D'Urso^{c,2}

^a Department of Educational Sciences, University of Catania, Via Biblioteca 4, Catania (CT), 95124, Italy

^b Laboratory of Cognitive and Social Neuroscience, Department of Cognitive Science, Psychology, Education and Cultural Studies (COSPECS), University of Messina, Messina, Italy

^c Independent Researcher, Tilburg, Netherlands

ARTICLE INFO

Keywords:

Pseudo-factor analysis
Pretrained language models
Psychometrics

ABSTRACT

The integration of Pretrained Language Models (PLMs) into psychometrics introduces new opportunities for scale development and validation. Pseudo-Factor Analysis (PFA), which employs item-level embeddings as proxies for human responses, offers a data-efficient alternative to Exploratory Factor Analysis (EFA). This study examines PFA's capacity to replicate EFA outcomes across clinical and health psychology measures in English and Italian. PFA showed high performance in English (factor recovery up to 100%, congruence >.90), while Italian results were more variable. Findings highlight both the promise and language sensitivity of PFA, underscoring its potential for cross-linguistic psychometric applications.

1. Introduction

Measurement is a cornerstone of psychological research, with the robustness of psychological measures typically evaluated through two key criteria: reliability and validity (Kline, 2015).

The development of reliable and valid psychological measures begins with the precise and operational definition of a psychological construct. This definition guides the creation of questions, or *items* that capture specific aspects of the construct (Furr and Bacharach, 2019). Given the variety of ways items can be formulated, researchers usually draft multiple versions and conduct pilot testing with a sample population to assess their psychometric properties. These properties are often analyzed using frameworks like Classical Test Theory (CTT) or Item Response Theory (IRT). CTT focuses on total test scores and provides indices such as reliability coefficients and measurement error, offering a straightforward evaluation of a test's overall quality. In contrast, IRT models the probability of a response as a function of both the respondent's underlying trait level and item characteristics (e.g., difficulty, discrimination), allowing a more fine-grained analysis of how individual

items perform across different populations.

Initial item sets, however, rarely achieve the desired level of quality, necessitating iterative refinements. Through repeated testing, researchers select and revise items with the aim of improving their reliability (e.g., stability over time), construct validity (i.e., extent to which items represent the intended construct) and external validity (i.e., the extent to which construct relates appropriately to others). After multiple refinements, the measure may eventually be considered suitable for broader applications.

Despite its rigor, this traditional process has notable limitations—most significantly; it is time-consuming and resource-intensive. Delay often stems from the practical demands of item development, study design, participants' recruitment, and data analysis.

Participants' recruitment can be costly. For example, using MTurk, a widely used crowdsourcing platform, the average payment rate is \$10.41 per hour (Moss et al., 2023). Developing a 100-item scale may require testing 300 preliminary items, with each participant spending approximately 20 min per session. To achieve appropriate statistical reliability, at least five participants per item is typically needed,

* Corresponding author.

E-mail addresses: simone.varrasi@unict.it (S. Varrasi), alessio.platania@unict.it (G.A. Platania), sabrina.castellano@unict.it (S. Castellano), francesca.ferraioli@unime.it (F. Ferraioli), simona.massimino@unime.it (S. Massimino), carmelomario.vicario@unime.it (C.M. Vicario), s.dinuovo@unict.it (S. Di Nuovo), dursodamiano@gmail.com (E.D. D'Urso).

¹ Simone Varrasi and Giuseppe Alessio Platania contributed equally to this work and are joint first authors.

² Epifanio Damiano D'Urso and Santo Di Nuovo are joint co-last authors.

<https://doi.org/10.1016/j.metip.2026.100244>

Received 2 October 2025; Received in revised form 3 March 2026; Accepted 8 March 2026

Available online 10 March 2026

2590-2601/© 2026 The Authors. Published by Elsevier Ltd. This is an open access article under the CC BY license (<http://creativecommons.org/licenses/by/4.0/>).

resulting in over 500 h of testing and costs exceeding \$5000.

Given these challenges, methods that can accelerate scale development are highly valuable. Among emerging approaches, Artificial Intelligence (AI)—particularly Pretrained Language Models (PLMs)—shows great promise for streamlining psychometric research, offering the potential to reduce time and cost while enhancing measurement quality.

1.1. Traditional exploratory factor analysis and factor recovery

In conventional psychometric practice, Exploratory Factor Analysis (EFA) represents the primary method for initially investigating the latent structure underlying a set of questionnaire items (Kline, 2015; Furr and Bacharach, 2019). EFA operates on the empirical covariance or correlation matrix and aims to reproduce this matrix through a smaller number of latent variables that account for the shared variance among items. The resulting solution is typically interpreted through the pattern matrix, which contains factor loadings reflecting the strength and direction of the relationship between each item and the latent factors (Lorenzo-Seva & ten Berge, 2006).

A central concern in EFA is factor recovery, that is, the extent to which the extracted solution approximates the true underlying structure of the construct. Recovery is influenced by multiple elements, including the quality of the correlation matrix, sample size, communality estimates, and the choice of extraction and rotation methods (Kiers, 1994). Rotations—whether orthogonal or oblique—are applied to achieve a more interpretable simple structure, with oblique rotations generally preferred when factors are expected to correlate, as is typical for psychological dimensions (Lorenzo-Seva et al., 2011).

The correspondence between an obtained factor solution and a theoretical or reference structure is commonly evaluated using indices such as Tucker's congruence coefficient (Tucker, 1951), which quantifies the similarity between loading patterns across solutions. These concepts—correlation matrices, factor loadings, rotation, and recovery—constitute the conceptual backbone that Pseudo-Factor Analysis (PFA) seeks to emulate by replacing the empirical response correlation matrix with a matrix of embedding-based semantic similarities (Guenole et al., 2025). In this sense, PFA can be understood as an EFA conducted on a linguistically derived representation of items rather than on human response data.

1.2. AI impact on the psychometric field

Artificial Intelligence (AI) is significantly impacting various fields, and its application to psychological research is on the rise (Innes and Morrison, 2021). Across different domains, AI technologies have been applied to support daily activities, (Petrich et al., 2022), workplace well-being (Lopes et al., 2023), and assessing both motor and cognitive abilities (Agrafiotis et al., 2021; Varrasi et al., 2019), as well as to the early detection of cognitive decline from speech patterns (Toth et al., 2018), and to the simulation and analysis of cognitive and behavioral processes (Kim and Kim, 2020). More broadly, machine learning methods are increasingly used to enhance data processing and ecological validity in psychological research (Orrù et al., 2020).

In the field of psychometrics, AI has introduced innovative methods for developing psychological measures, particularly using Pretrained Language Models (PLMs)—computational models trained for language processing tasks (Arnulf et al., 2024). This advancement is largely driven by the creation of language embeddings—high-dimensional vectors that numerically represent text and can estimate psychometrically relevant parameters (Hassani and Silva, 2023). Built upon deep neural networks and transformer architectures (Vaswani et al., 2017), these embeddings address the limitations of earlier closed language models (Tausczik and Pennebaker, 2010) by incorporating contextual word associations, closely aligning with Natural Language Processing (NLP) methodologies.

This approach rests on the theoretical assumption that semantic similarity between item texts may approximate patterns of response covariance observed in human data. This assumption is grounded in principles from distributional semantics, according to which linguistic meaning emerges from patterns of use and contextual co-occurrence (Harris, 1954; Landauer and Dumais, 1997). From a psychometric perspective, questionnaire items designed to measure the same latent construct are expected to share overlapping semantic content, reflecting similar cognitive interpretations and response tendencies (Tourangeau et al., 2000).

Accordingly, embedding-based representations may capture aspects of the conceptual proximity between items that underline shared variance in human responses. However, this correspondence is probabilistic rather than deterministic and is expected to hold more strongly for constructs with clear, stable semantic cores than for those relying on experiential, affective, or context-dependent interpretations. From the standpoint of cognitive models of item responding, responses to self-report measures involve the interpretation of item wording, mapping onto internal representations, and selection of a response option. Semantic similarity between items may therefore reflect shared cognitive processing demands and overlapping interpretative frames, which in turn contribute to correlated response patterns at the latent level.

Although early concerns regarding reliability, validity, and biases—issues shared by both AI and traditional psychometrics—persist, recent studies increasingly highlight the potential of PLMs for psychological measurement (Landers and Behrend, 2022).

PLMs have been integrated into psychometrics for various purposes, including item generation, item crowd-responding, and semantic-based refinement. Research shows that items generated by PLMs perform comparably to those created by humans (Laverghetta and Licato, 2023). Similarly, questionnaires completed by artificial respondent groups yield psychometric properties similar to those from actual participant samples (Laverghetta et al., 2022). Furthermore, PLMs enable the enhancement of psychometric properties by analyzing the semantic structures within psychological measures (Hernandez and Nie, 2023).

From a psychometric perspective, the most relevant aspect of PLMs is not their general AI capabilities but their ability to generate sentence embeddings that encode the semantic content of questionnaire items in continuous vector spaces (Reimers and Gurevych, 2019). When two items express similar psychological meaning, their embeddings tend to be closer, mirroring the mechanism through which respondents produce correlated answers (Tourangeau et al., 2000).

Recent work has therefore investigated whether embedding similarity can approximate empirical covariance structures and support inferences about dimensionality, cross-loadings, and factor correlations prior to data collection (Guenole et al., 2025; Wulff and Mata, 2024; Feraco and Toffalini, 2025). From this viewpoint, PLM-based approaches do not replace psychometric theory but provide an alternative representation of the same constructs, grounded in distributional semantics (Harris, 1954; Landauer and Dumais, 1997).

Consequently, the integration of embeddings within psychometric workflows can be conceived as a complementary strategy that leverages linguistic information to anticipate structural properties of measurement models, while maintaining the conceptual and methodological framework of traditional factor analysis.

Given the high costs of traditional validation studies, PLMs offer an efficient alternative through methods such as pseudo-item discrimination, which approximates empirical measurement by using semantic similarity rather than participant data (Guenole et al., 2025). This approach relies on language embeddings that capture semantic information to account for item variance. Specifically, item-to-construct correlations are estimated by comparing an item's semantic content with the PLM's representation of the construct (Wulff and Mata, 2024). In doing so, this method helps exclude semantically irrelevant items that may statistically fit a model but fail to represent the construct accurately (Babcock and Albano, 2012).

Nonetheless, pseudo-item discrimination has significant limitations. Because it evaluates items against constructs one scale at a time, it cannot account for multidimensional structures, (Guenole et al., 2025). To address this issue, researchers have developed Pseudo-Factor Analysis (PFA)—a technique designed to estimate latent construct parameters, including dimensionality, factors, and indicators, without relying on response data (Guenole et al., 2025). PFA operates under the assumption that the empirical covariance matrix of item responses can be approximated by a cosine similarity matrix derived from item embeddings. This substitution enables researchers to explore expected model parameters even prior to data collection.

1.3. Current limitations of pseudo-factor analysis approach

Despite its advantages, several limitations of Pseudo-Factor Analysis (PFA) remain evident in the literature. To date, PFA has primarily been tested on personality scales (Guenole et al., 2025). While initial results are promising, further research is necessary to evaluate its effectiveness across a broader range of psychometric measures, including those assessing clinical symptoms. Another key limitation is the language dependency of existing studies. Thus far, PFA has only been applied to measurement tools developed in English, raising concerns about its cross-cultural validity. Expanding research to include measures in other languages is essential to confirm the generalizability of PFA across diverse linguistic and cultural contexts.

Additionally, PFA has predominantly been applied to facet-level measurement models, such as the Revised NEO Personality Inventory (NEO-PI-R), which decomposes each of the Big Five domains into six specific facets (Costa and McCrae, 1992). However, many psychometric scales lack this three-level structure, directly estimating factors from individual items. This structural difference may affect PFA's applicability and effectiveness across a wider range of measurement models.

In addition to these structural and linguistic limitations, it is important to acknowledge that PFA fundamentally relies on text embeddings, which are not neutral representations of psychological constructs. Item embeddings are shaped by multiple sources of variance, including the characteristics of the training corpora, fine-tuning objectives, architectural constraints of the language models, and the semantic properties of the input text. In cross-language settings, further sources of distortion may arise from translation quality, semantic shifts between languages, and differential cultural salience of construct-related terminology. As a result, similarity structures derived from embeddings may reflect not only latent psychological dimensions, but also model-specific and language-specific semantic regularities. These factors must be explicitly considered when interpreting PFA results, particularly in multilingual applications.

Given PFA's potential as an innovative scale development tool, addressing these limitations is crucial. Expanding research to non-English scales and various psychometric models will help determine whether PFA yields results comparable to traditional psychometric methodologies. Such validation would strengthen confidence in PFA as a reliable and efficient assistant for scale development, potentially streamlining the validation process while maintaining scientific rigor.

Recent studies, such as Wulff and Mata (2024) on semantic overlap of constructs, and Feraco and Toffalini (2025) on SEMbeddings, highlight both the opportunities and challenges of using embeddings in psychometrics, reinforcing the need of evaluating PFA across different languages and clinical instruments.

These unresolved limitations directly motivate the present study, which examines whether PFA can be meaningfully extended beyond English-language personality scales.

1.4. Study goals & manuscript overview

The present study is conceived as an exploratory and comparative investigation rather than a formal validation study. Its primary goal is to

examine whether Pseudo-Factor Analysis (PFA) yields factor solutions that meaningfully correspond to those obtained through traditional empirical approaches when applied beyond English-language personality scales.

Specifically, we evaluate the degree of convergence between PFA-derived structures and previously published EFA/Exploratory Structural Equation modeling (ESEM) results across health and clinical instruments in English and Italian. By doing so, the study aims to assess under which linguistic and methodological conditions PFA can provide informative approximations of key model characteristics, such as dimensionality and relative item-factor associations, and where its limitations become evident.

In this sense, the contribution of the present work lies in delineating the boundaries of applicability of PFA, rather than in establishing it as a substitute for empirical psychometric validation or clinical deployment.

It is also important to note that in this work, we rely on pretrained encoder-only language models and sentence transformers (e.g., DistilRoBERTa, MiniLM, MPNet) to obtain fixed-length vector representations of questionnaire items. For clarity, and in line with current terminology, we refer to these models as pretrained language models or sentence encoders, rather than large language models (LLMs), as the present approach exclusively relies on pretrained text embeddings and does not involve generative or reasoning-based capabilities.

2. Materials and methods

2.1. Data

To evaluate the performance of PFA on health/clinical and non-English psychological measures, we established specific criteria for selecting appropriate psychometric scales. The selection criteria were as follows: First, we selected measures available in both Italian and English, with validated versions in each language where possible. Second, we focused on scales that had a high applicability to key psychological constructs or frequent use in practice. Third, we only selected scales where we could get full access to all scale items. Fourth, we selected studies with prior validation using either Exploratory Factor Analysis (EFA) or Exploratory Structural Equation modeling (ESEM). Finally, we only looked at studies with detailed reporting of EFA results, including the complete factor loading matrix, in the original validation studies. Based on these criteria, three well-established scales were selected: Depression, Anxiety, Stress Scales (DASS-21), and Dark Triad Dirty Dozen (DTDD) and Mesure du Stress Psychologique (MSP).

The Depression, Anxiety, Stress Scales (Bottesi et al., 2015) is a widely used clinical assessment tool designed to measure three core pathological dimensions, namely: Depression, Anxiety, and Stress. The 21 items in the DASS are rated on a 4-point Likert scale. Then, the Dark Triad Dirty Dozen (Jonason and Webster, 2010) assesses three socially aversive but distinct personality traits, collectively referred to as the Dark Triad, that is, Machiavellianism (*manipulative behavior*), Narcissism (*excessive self-love*), and Psychopathy (*lack of empathy*). The scale consists of 12 statements, each rated based on the respondent's level of agreement. The DTDD is often utilized in studies exploring antisocial behaviors and personality dynamics. Finally, we also selected the MSP (Di Nuovo et al., 2000), which is a self-report questionnaire comprising 49 items rated on a 4-point Likert scale. It is designed to identify dysfunctional stress by evaluating various factors, including loss of control and irritability, psychophysiological sensations, sense of effort and confusion, depressive anxiety, pain and physical problems, hyperactivity and accelerated behaviors. The MSP is grounded in a theoretical framework that views stress as a natural, adaptive response—functional within certain thresholds but potentially dysfunctional when exceeding them.

The MSP is the only non-English validated scale, which we included to evaluate (i) if PFA results could generalize to unidimensional scales and (ii) non-native English contexts. In fact, unlike the DASS-21 and

DTDD, which required translation into Italian, the MSP was originally developed and validated in French (Lemyre and Tessier, 1988) and subsequently translated into Italian but not English. As such, it enables the assessment of whether PFA results generalize beyond English-origin instruments and whether PLMs—often trained on English-dominant corpora—exhibit differential performance based on the scale's original language.

2.2. Procedure

The analytical pipeline implemented in this study followed a multi-step approach similar to that in Guenole et al. (2025) to evaluate the performance of Pseudo-Factor Analysis (PFA) on health, clinical, and non-English psychological measures. The procedure consisted of five primary steps.

The process began with item text collection, where items from the validation studies of each selected scale were gathered in both Italian and English. To maintain fidelity to the original instruments, the exact wording from the original validation studies was preserved. Each item was categorized based on the corresponding factor identified in the validation study, and we recorded whether it was positively or negatively worded.

The next step involved PLM selection. To generate item embeddings, we used a range of sentence encoder models available on Hugging Face (*n.d.*). In selecting these models, priority was given to those utilized in prior psychometric research, particularly those referenced by Guenole et al. (2025), to maintain methodological continuity. The models, both English and multilingual, are displayed in Table 1. For this task we decided to only use sentence transformers instead of simpler architectures (e.g., BERT, GloVe) as the former have been shown to be better suited for sentence classification tasks as well as sentence-based semantic similarity calculations (Reimers and Gurevych, 2019). Further, we selected only open models since they allow for more customization, fine-tuning, and provide greater transparency, allowing further research and analysis. Moreover, we picked models used in Guenole et al. (2025) study, with large number of downloads on huggingface and, wherever possible, that are available both in an English-only and multilingual version (i.e., mpnet). Notably, the Wulff and Mata (2024) model, fine-tuned specifically on personality data, was included for its strong performance in personality assessments, despite its lack of validation in clinical contexts.

Following model selection, we proceeded with cosine similarity calculation. Sentence embeddings were generated for each item using the selected PLMs. To evaluate if multiple-model information could be beneficial, average embeddings were also computed by aggregating the outputs from all models. Using these embeddings, a cosine similarity matrix was constructed. The diagonal of the matrix represented the cosine self-similarity of each item, while the off-diagonal values indicated pairwise similarities between different items. This matrix was subsequently treated as a correlation matrix, serving as the foundation

for factor analysis.

The fourth stage involved conducting an Exploratory Factor Analysis (EFA), followed by target rotation. Since PFA fundamentally operates as an EFA applied to PLM-derived embeddings, we followed canonical EFA procedures. We extracted a number of factors consistent with those identified in the original validation studies, ensuring a direct comparison between the PFA results and traditional EFA findings. Importantly, no analyses were conducted to explore alternative factor structures, as sample size considerations—central to such model comparisons—are not applicable in PFA. For the rotation process, we applied a fully specified oblique target rotation, an informed method that minimizes the distance between the observed unrotated loading matrix and a user-defined target matrix. This approach facilitated the direct comparability of our PFA results with those obtained in prior validation studies. However, any type of rotation that is applicable to EFA can, in principle, be implemented with PFA. For instance, one may use uninformed rotation approaches (e.g., oblimin, varimax) as well as rotation approaches that automatically locate an optimal semi-specified target rotation (e.g., simplimax; Kiers, 1994).

The final step in the analytical pipeline was factor labelling. Determining the interpretability of factors is a critical step in EFA, and this complexity extends to PFA. To ensure comparability with existing validation studies, we employed the Dominant Average Absolute Loading (DAAL) method, as outlined by Guenole et al. (2025). This approach involved calculating the average absolute loading of item groups for each factor, based on the factor structure reported in the original validation studies. The factor was then labeled according to the item group with the highest DAAL score. For instance, in the DTDD validation study, the first 4 items (“I tend to manipulate others to get my way”, “I have used deceit or lie to get my way”, “I have used flattery to get my way” and “I tend to exploit others towards my own end”) were found to be those with the highest loadings on the machiavellianism factor, items 5 to 8 primarily loaded on the psychopathy factor and the last 4 items on the narcissism factor. For each of these three groups, we calculated the average loading on each factor. Then, we labeled a factor according to which group of items had the highest DAAL score on that factor. In cases where multiple sets of items were a factor that had none or multiple unique sets of items to it, we considered that factor unassigned.

2.3. Measures

To evaluate the performance of Pseudo-Factor Analysis (PFA), we employed two primary metrics, following the methodology outlined by Guenole et al. (2025): Factor Recovery Rate and Factor Congruence.

The Factor Recovery Rate was used to assess the proportion of successfully recovered factors across various encoding methods for both the DTDD and the DASS-21. For the DTDD, we used the items and empirical results from Schimmenti et al. (2019) and Gomez et al. (2020) for the Italian and English validation, respectively. For the DASS-21, we used

Table 1
Pretrained language models used for pseudo-factor analysis.

Model Name	Output Token length	Maximum number of tokens	Source Architecture	Key reference	Huggingface nomenclature	Language capability
DistilRoberta	768	75	Roberta	Reimers and Gurevych (2019)	nli-distilroberta-base-v2	Multilingual
mpnet	768	128	XLNet	Reimers and Gurevych (2019)	paraphrase-multilingual-mpnet-base-v2	Multilingual
miniLM	768	128	Bert	Reimers and Gurevych (2019)	paraphrase-multilingual-miniLM-L12-v2	Multilingual
e5	768	512	XLNet	Wang et al. (2024)	paraphrase-multilingual-e5-base	Multilingual
W&M	768	384	XLNet	Wulff and Mata (2024)	mpnet-personality	Multilingual
mpnet en	768	128	XLNet	Reimers and Gurevych (2019)	all-mpnet-base-v2	English
t5	768	128	Text-to-Text transfer	Ni et al. (2022)	sentence-t5-base	English

the items and empirical results from Iannattone et al. (2024), and Jon-ason and Webster (2010), for the Italian and English validation, respectively. This metric was not applied to the MSP, for which we used item text and validation results from Di Nuovo et al. (2000), considering the test as a strong unidimensional measure, as supported by the international literature (Fillion et al., 1989; Lemyre and Lalande-Markon, 2009).

Factor Congruence, on the other hand, evaluated the degree of similarity between the factor loadings obtained through PFA and those reported in the original validation studies. To quantify this similarity, we employed Tucker's congruence coefficient (Tucker, 1951), a well-established statistical measure for factor comparison. The interpretative thresholds for this coefficient follow conventional standards: values above 0.85 indicate fair similarity, while values of 0.95 or higher denote excellent similarity (Lorenzo-Seva & ten Berge, 2006).

2.4. Cross-language comparison

An essential component of this study was to examine whether PLMs performance in PFA is language dependent. To this end, we compared factor recovery and congruence results for both the original English versions of the scale items and their Italian translations.

By conducting parallel analyses across these two language versions, we assessed the extent to which PLMs preserve the integrity of the measurement model structures when applied to translated items. This cross-language comparison is particularly relevant for assessing the transcultural applicability of PFA, especially given that psychometric tools often undergo linguistic adaptation prior to broader use. The inclusion of the MSP, originally developed in French, further enriched this analysis by providing a not native English reference point.

In more detail, we used the Italian translations provided in the validation studies of the psychometric tools: the DASS-21 in Bottesi et al. (2015), the DTDD in Schimmenti et al. (2019), and the MSP in Di Nuovo et al. (2000).

This comparative approach allowed us to determine whether the semantic processing capabilities of PLMs remain consistent across languages or if translation introduces distortions that affect factor recovery and congruence outcomes.

2.5. Software and open practices

Item embeddings were computed using python version 3.10 (Van Rossum and Drake, 2009), while psychometric analyses were conducted in R version 4.3.1 (Moustafa et al., 2020). We specifically utilized sentence transformers version 3.4.1 (Reimers and Gurevych, 2019) to calculate sentence embeddings, readxl version 1.4.3 (Wickham et al., 2019) for files reading, factoMineR version 2.11 (Lé et al., 2008) for target-rotated factor analysis and psych version 2.4.6 (Revelle, 2017) for factor congruence metrics. Our entire analytical scripts and data are freely available at <https://osf.io/p2f5e/>

3. Results

3.1. English version

Tables 2 and 3 display the factor recovery and congruence results for the scales' English versions, respectively. DistilRoberta, MiniLM, MPNet (EN), T5, and the Wulff & Mata model achieved a 100% factor recovery rate, correctly identifying all factors across both scales. However, both the MPNet and E5 models demonstrated reduced performance, with a 75% recovery rate, failing to recover some factors, particularly in the DTDD. The Depression and Psychopathy factors were the most consistently recovered across all models, suggesting that these constructs are well-represented semantically within the PLM embeddings. In contrast, the Anxiety and Stress factors from the DASS-21 were less consistently identified, particularly by the E5 model.

Most models achieved congruence values above the accepted cut-off for good similarity (0.85), indicating that PFA successfully replicated traditional factor structures in most cases. The T5 and DistilRoberta models consistently achieved the highest congruence scores (both exceeding 0.90 across most factors), indicating strong alignment with the original factor structures. However, the multilingual MPNet and the Wulff & Mata model showed reduced congruence for certain factors, particularly for the Anxiety and Stress dimensions in the DASS-21, which yielded scores below the 0.85 threshold.

3.2. Italian version

Tables 4 and 5 display the factor recovery and congruence results for the scales' Italian versions, respectively. The Italian version exhibited more variability in factor recovery compared to the English version. The MiniLM and MPNet models achieved 100% factor recovery, successfully identifying all expected factors across both the DASS-21 and DTDD. Conversely, the DistilRoberta, Wulff & Mata, and E5 models underperformed, with recovery rates ranging between 33% and 66%. Only the Depression and Psychopathy factors were consistently recovered above the 80% threshold, mirroring the results found in the English version. In contrast, the Anxiety and Stress factors from the DASS-21 exhibited the lowest recovery rates, suggesting that translation may compromise the structural integrity of these constructs.

Congruence analysis revealed moderate success in replicating factor structures in the Italian versions. While some models, notably MPNet and E5, achieved congruence values above the 0.85 cut-off for several factors, other models struggled to maintain consistency across all constructs. The highest congruence was observed for the Depression and Psychopathy factors, while Anxiety and Stress once again exhibited the lowest congruence scores. Interestingly, the MSP scale, which was originally developed in French and then translated into Italian, displayed near-perfect congruence across all models (0.97), suggesting that PFA may be able to handle non-English original scales.

Table 2
Factor recovery English scales.

Model	DASS-21			DTDD			Mean model
	Depression	Anxiety	Stress	Machiavellianism	Psychopathy	Narcissism	
DistilRoberta	1	1	1	1	1	1	100%
miniLM	1	1	1	1	1	1	100%
mpnet	1	1	1	0	1	0	75%
wulff	1	1	1	1	1	1	100%
e5	1	0	0	1	1	1	75%
mpnet_en	1	1	1	1	1	1	100%
t5	1	1	1	1	1	1	100%
Avg. Transformer	1	0	0	1	1	1	75%
Mean Factor	100%	75%	75%	87.5%	100%	87.5%	

Table 3
Factor congruence English scales.

Model	DASS-21			DTDD			Mean model
	Depression	Anxiety	Stress	Machiavellianism	Psychopathy	Narcissism	
DistilRoberta	0.92	0.88	0.87	0.96	0.95	0.95	0.92
miniLM	0.87	0.87	0.81	0.94	0.80	0.91	0.87
mpnet	0.90	0.77	0.54	na	0.84	na	0.76
wulff	0.94	0.84	0.52	0.88	0.91	0.87	0.83
e5	0.92	na	na	0.91	0.94	0.84	0.90
mpnet_en	0.94	0.79	0.81	0.93	0.94	0.94	0.89
t5	0.95	0.89	0.85	0.92	0.97	0.96	0.92
Avg. Transformer	0.94	na	na	0.94	0.95	0.96	0.95
Mean Congruence	0.92	0.84	0.73	0.93	0.91	0.92	

Table 4
Factor recovery Italian scales.

Model	DASS-21			DTDD			Mean model
	Depression	Anxiety	Stress	Machiavellianism	Psychopathy	Narcissism	
DistilRoberta	1	0	0	0	1	0	33%
miniLM	1	1	1	1	1	1	100%
mpnet	1	1	1	1	1	1	100%
wulff	0	1	0	0	1	0	33%
e5	1	0	0	1	1	1	66%
Avg. Transformer	1	1	1	0	1	0	66%
Mean Factor	83%	66%	50%	50%	100%	50%	

Table 5
Factor congruence Italian scales.

Model	DASS-21			DTDD			MSP	Mean model
	Depression	Anxiety	Stress	Machiavellianism	Psychopathy	Narcissism		
DistilRoberta	0.73	na	na	na	0.87	na	0.97	0.86
miniLM	0.86	0.69	0.75	0.88	0.87	0.87	0.97	0.84
mpnet	0.94	0.75	0.78	0.94	0.86	0.87	0.97	0.87
wulff	na	0.79	na	na	0.80	na	0.96	0.85
e5	0.87	na	na	0.92	0.92	0.93	0.97	0.92
Avg. Transformer	0.91	0.77	0.75	0.83	0.83	na	0.97	0.84
Mean Factor Congruence	0.86	0.75	0.76	0.91	0.85	0.89	0.97	0.85

3.3. General findings

Figs. 1 and 2 present the bar charts of factor recovery and Tucker's congruence, providing a direct visual comparison of PFA performance across languages and models. These visualizations complement the numerical results reported in Tables 2–5 and support the interpretation of cross-language differences in dimensional recovery and structural similarity. As shown in Fig. 1, factor recovery was consistently high for the English versions across most models, with several encoders achieving full recovery of all dimensions. In contrast, the Italian versions exhibited

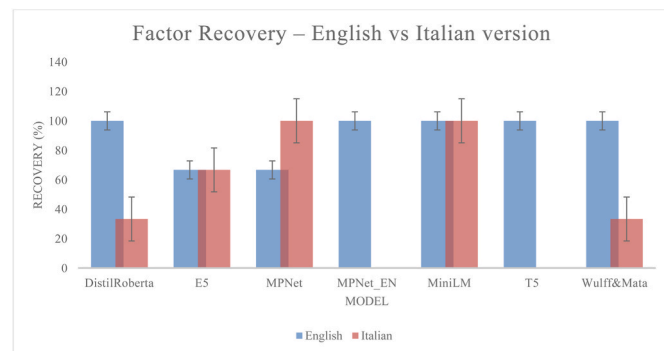


Fig. 1. Factor recovery across models for English vs. Italian versions. Values represent the percentage of factors correctly recovered by PFA.

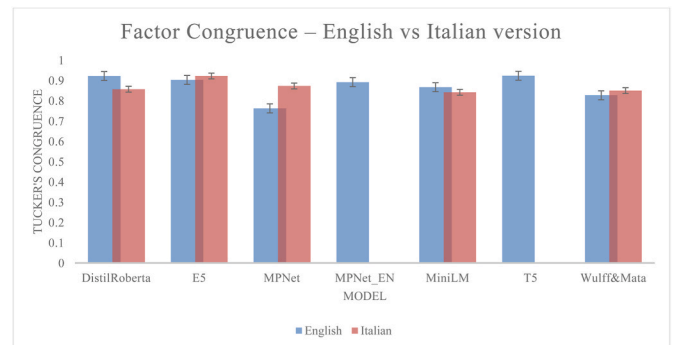


Fig. 2. Tucker's congruence across models for English vs. Italian versions. Values represent the similarity of factor structures (>.85 = fair; >.95 = excellent).

greater variability, particularly for the Anxiety and Stress factors of the DASS-21, whereas Depression and Psychopathy remained the most stable dimensions across languages.

Fig. 2 further illustrates that Tucker's congruence followed a similar pattern: English instruments generally reached values above the .85 threshold, while Italian translations showed more heterogeneous results, except for the MSP scale, which displayed near-perfect congruence across all models.

For the English versions of the scales, PFA was highly effective in replicating traditional factor structures. Most models, particularly T5, DistilRoberta, MiniLM, and MPNet (EN), achieved near-perfect factor recovery rates and high factor congruence, with Tucker's congruence coefficients frequently exceeding the 0.85 threshold for good similarity. Depression and Psychopathy factors were consistently well-recovered across all models, indicating strong semantic alignment with PLM embeddings. However, challenges emerged with the Anxiety and Stress factors from the DASS-21, where models such as E5 and Multilingual MPNet, struggled to reach acceptable congruence levels.

In contrast, the Italian versions of the scales showed greater variability in results. While MiniLM and MPNet achieved high factor recovery rates comparable to their performance on the English scales, other models, including DistilRoberta and Wulff & Mata, showed a notable decline in both recovery and congruence. Depression and Psychopathy factors *remained* the most reliably recovered, while Anxiety and Stress factors demonstrated reduced recovery rates and lower congruence, reflecting the patterns similar to those observed in the English versions.

A significant observation was the consistently high performance on the MSP scale, originally developed in French and then translated into Italian. All models demonstrated near-perfect congruence (>0.95) for this scale, suggesting that PLM-based PFA may be able to handle non-English original scales. This outcome may also reflect the relative simplicity of MSP's unidimensional structure, which facilitates high congruence. Further, when compared to the lower results of Italian scales adapted from English, this result may suggest that PLMs show differential performance in capturing semantic similarities when a corresponding English source exists. As an example of the loading's matrices obtained for the different languages, we display the empirical and mpnet loadings for the English and Italian items of DASS-21 scale in [Tables 6 and 7](#), respectively.

Overall, the findings indicate that while PFA, when coupled with PLMs, is highly effective in replicating traditional factor structures in English psychometric tools, its performance in non-English contexts is more variable, potentially influenced by the quality of translations and the inherent language-dependence of PLMs. These results are consistent with other recent applications of embeddings in psychometrics (e.g., [Wulff and Mata, 2024](#); [Feraco and Toffalini, 2025](#); [Jung and Seo, 2025](#)), which similarly reported strong replication of factor structures in English-language scales but greater variability in translated instruments.

4. Discussion

This study aimed to compare the outcomes of Pseudo-Factor Analysis (PFA) with those of traditional Exploratory Factor Analysis (EFA), assessing the ability of PLM-based item embeddings to replicate established psychometric structures across both English and Italian versions of well-validated psychological instruments. Given PFA's potential to accelerate psychometric scale development by reducing the need for extensive data collection and stepwise validation, the study provides valuable insights into its practical strengths and limitations.

Building on pioneering work in the field ([Guenole et al., 2025](#)), this study extends the application of PFA to non-English psychometric tools within the domains of health and Clinical Psychology. By including the Mesure du Stress Psychologique (MSP), the Depression, Anxiety, Stress Scales (DASS-21), and the Dark Triad Dirty Dozen (DTDD), we explored PFA's adaptability across various psychological domains and linguistic contexts.

Overall, results indicate that PFA effectively recovers and replicates factorial structures in English-language instruments, particularly for well-defined constructs like Depression and Psychopathy. Performance was less consistent for Italian translations, with greater variability in factor recovery and congruence. These findings suggest that PFA's *effectiveness* is influenced by both linguistic factors and the complexity of the constructs under examination. Indeed, as illustrated in [Figs. 1 and 2](#),

Table 6
English empirical and mpnet loading matrices of DASS-21 scale.

Item	Empirical English			Embeddings mpnet_en		
	D	A	S	D	A	S
I couldn't seem to experience any positive feeling at all	.74	-.21	.24	0.50	0.16	0.22
I found it difficult to work up the initiative to do things	.44	.09	.19	0.56	−0.23	0.61
I felt that I had nothing to look forward to	.80	.03	-.03	0.91	−0.12	0.14
I felt downhearted and blue	.74	.09	-.02	0.57	0.20	−0.01
I was unable to become enthusiastic about anything	.75	-.11	.16	0.72	−0.23	0.41
I felt I wasn't worth much as a person	.80	.09	-.08	0.90	−0.27	0.12
I felt that life was meaningless	.92	00	-.15	0.95	−0.15	0.03
I was aware of dryness of my mouth	.08	-.14	.58	−0.01	0.54	0.24
I experienced breathing difficulty (e.g. excessively rapid breathing, breathlessness in the absence of physical exertion)	-.04	.46	.36	−0.27	0.51	0.34
I experienced trembling (e.g. in the hands)	.09	.38	.12	−0.27	0.80	0.36
I was worried about situations in which I might panic and make a fool of myself	.20	.58	.02	−0.07	0.56	0.46
I felt I was close to panic	.13	.66	.12	0.00	0.79	0.17
I was aware of the action of my heart in the absence of physical exertion (e.g. sense of heart rate increase, heart missing a beat)	-.10	.47	.34	−0.13	0.51	0.22
I felt scared without any good reason	.17	.57	.10	0.20	0.67	0.09
I found it hard to wind down	.06	-.33	.41	0.19	0.17	0.62
I tended to over-react to situations	.14	.36	.32	0.28	0.18	0.55
I felt that I was using a lot of nervous energy	.08	.49	.33	−0.13	0.77	0.55
I found myself getting agitated	.20	.29	.31	0.15	0.41	0.48
I found it difficult to relax	.20	.38	.24	0.27	0.26	0.68
I was intolerant of anything that kept me from getting on with what I was doing	.30	.00	.30	0.52	−0.18	0.41
I felt that I was rather touchy	.18	.21	.36	0.24	0.30	0.28

the comparison between English and Italian versions revealed systematic differences in both factor recovery and Tucker's congruence, with greater variability for translated instruments and more stable performance for constructs with clearer semantic cores.

The contrast between the English and Italian versions underscores the language dependency of PLM-based PFA. While the method performed well with native English instruments, challenges emerged with translated scales, particularly for constructs like Anxiety and Stress, which exhibited lower recovery rates and reduced congruence. This highlights a key limitation of current PFA applications: their sensitivity to semantic shifts introduced during translation.

Importantly, the strong performance of the MSP in Italian, also evident in [Fig. 2](#) and despite its original development in French, suggests that the conceptual coherence and construct validity of the original instrument may play a more critical role than the target language per se. This finding points to a distinction between surface-level linguistic translation and deeper cultural and conceptual alignment of the measured construct.

These findings also underscore an important conceptual implication:

Table 7

Italian empirical and mpnet loading matrices of DASS-21 scale.

Item	Empirical Italian			Embeddings mpnet		
	D	A	S	D	A	S
Non riuscivo proprio a provare delle emozioni positive	.51	-.07	.28	0.74	0.06	0.13
Ho avuto un'estrema difficoltà nel cominciare quello che dovevo fare	.37	-.03	.12	0.20	0.39	0.40
Non vedo nulla di buono nel mio futuro	.76	.02	.03	0.88	0.03	-.29
Mi sono sentito scoraggiato e depresso	.59	.02	.27	0.68	0.18	0.17
Non c'era nulla che mi dava entusiasmo	.70	-.17	.20	0.90	-.13	-.07
Sentivo di valere poco come persona	.71	.20	.17	0.75	0.09	-.12
Sentivo la vita priva di significato	.87	.01	.01	0.85	-.03	-.23
Mi sono accorto di avere la bocca secca	-.11	.54	.09	0.23	0.27	0.41
Mi sono sentito molto in affanno con difficoltà a respirare (per es. respiro molto accelerato, sensazione di forte affanno in assenza di sforzo fisico)	-.16	.86	-.28	0.06	0.42	0.54
Ho avuto tremori (per es. alle mani)	-.21	.83	-.02	-.03	0.51	0.56
Ho temuto di trovarmi in situazioni in cui sarei potuto andare nel panico e rendermi ridicolo	.11	.68	-.06	0.06	0.99	0.23
Ho sentito di essere vicino ad avere un attacco di panico	-.03	.81	-.24	-.09	0.83	0.47
Ho percepito distintamente il battito del mio cuore senza aver fatto uno sforzo fisico (per es. battito cardiaco accelerato o perdita di un battito)	-.18	.86	.02	0.10	0.25	0.34
Mi sono sentito spaventato senza ragione	.01	.68	.01	0.33	0.60	0.21
Ho provato molta tensione e ho avuto difficoltà a recuperare uno stato di calma	-.10	.36	.34	0.06	0.582	0.71
Ho avuto la tendenza a reagire in maniera eccessiva alle situazioni	-.01	.34	.67	0.09	0.63	0.41
Ho sentito che stavo impiegando molta energia nervosa	-.04	.19	.49	0.04	0.60	0.75
Mi sono sentito stressato	-.01	.28	.50	0.27	0.54	0.58
Ho avuto difficoltà a rilassarmi	.22	-.31	.46	0.09	0.38	0.72
Non riuscivo a tollerare per nulla eventi o situazioni che mi impedivano di portare avanti ciò che stavo facendo	.17	.22	.48	0.35	0.37	0.18
Mi sono sentito piuttosto irritabile	.01	.01	.80	0.36	0.41	0.52

PFA is inherently dependent on the properties of the embedding models used to generate item representations. Differences in factor recovery and congruence across models should not be interpreted solely as noise or instability, but rather as evidence that the embedding space itself constitutes a substantive component of the measurement process. In this sense, PFA is best understood not as a language-agnostic psychometric procedure, but as a model-dependent analytic lens whose outputs reflect

the interaction between item semantics, language, and the training characteristics of the underlying PLMs.

Importantly, the exceptional performance of the MSP—originally developed in French and later translated into Italian—suggests that PLM-based PFA can effectively handle non-English original scales. This finding challenges the assumption that native-language instruments are inherently easier for PLMs to process and instead points to more nuanced effects related to language origin and translation. The contrast between the MSP's high congruence (>0.95) and the lower congruence observed for the Italian translations of the DASS-21 and DTDD—both originally developed in English—suggests that PLMs may process scales differently depending on the presence of the original source language in their training data.

One plausible explanation for the weaker results in translated instruments is the well-documented *Anglocentric bias* of PLMs. Because these models are predominantly trained on English-language corpora, they tend to prioritize Anglo-American semantic structures and under-represent non-English nuances (Havaldar et al., 2023; Levisen, 2022). This limitation may partly explain why scales translated from English into Italian performed less consistently than the MSP, which originated in French. Beyond methodological considerations, the observed Anglo-centric bias in embedding-based representations raises broader ethical concerns for psychological research and assessment. The disproportionate performance of instruments originally developed in English reflects not only technical asymmetries in training data, but also longstanding inequities in how psychological constructs are defined, operationalized, and disseminated across cultural contexts.

From this perspective, reliance on pretrained language models risks reinforcing existing epistemic hierarchies, whereby constructs articulated in English are implicitly treated as normative, while non-English or culturally localized instruments are evaluated against Anglo-centric standards. Without explicit attention to these dynamics, embedding-based methods may inadvertently perpetuate structural biases in global psychological science rather than mitigate them.

Another key insight from this study is the significant impact of PLM selection on PFA performance. While models like T5, MiniLM, and MPNet (EN) consistently produced strong results, others, such as E5 and the multilingual version of MPNet, showed inconsistent performance across different scales and languages. This suggests that researchers must carefully select the models, and that fine-tuning PLMs on field-specific psychological corpora could improve the models' performance.

Beyond practical model selection considerations, the observed differences in performance across PLMs likely reflect fundamental differences in architecture, training objectives, and linguistic exposure. Sentence encoders vary substantially in the corpora used for pretraining, the degree of multilingual optimization, and the downstream tasks used for fine-tuning (e.g., natural language inference, paraphrase detection, or domain-specific semantic similarity). Models optimized for semantic similarity tasks in English-dominant datasets may produce embedding spaces that better capture construct-relevant semantic overlap for English items, whereas multilingual models must balance cross-linguistic alignment with representational specificity.

Furthermore, fine-tuned models such as the Wulff and Mata encoder may encode domain-relevant personality semantics that improve performance in some contexts while reducing generalizability to clinical constructs outside the fine-tuning domain. Similarly, differences in tokenization strategies and contextual representation mechanisms may influence how affective or experiential constructs—such as Anxiety and Stress—are represented in embedding space.

These factors suggest that variability across models should not be interpreted solely as methodological instability, but rather as evidence that PFA outcomes are partially contingent upon the epistemic assumptions embedded within pretrained language models. Future research would benefit from systematically comparing model characteristics, including multilingual alignment strategies and domain-specific fine-tuning, to better understand how embedding design

choices influence psychometric inference. Accordingly, the absence of a fully systematic theoretical framework for model comparison represents a limitation of the present study and should be addressed in future work.

The findings of this study carry several important implications for the future use of PFA and PLM-based item embeddings for psychometric research. First, the reduced performance observed in translated scales highlights the need for cross-lingual fine-tuning of PLMs to improve their ability to handle diverse linguistic contexts. This is particularly critical for psychometrics, where measurement instruments are frequently adapted for use in multiple languages. Second, the persistent difficulty in recovering certain constructs, such as Anxiety and Stress, suggests that some psychological dimensions may be semantically more complex or context dependent. Finetuning PLM could improve embeddings and, in turn, PFA's sensitivity to these nuances.

The persistent difficulty in recovering Anxiety and Stress factors suggests that PFA is more robust for semantically well-defined constructs (e.g., Depression, Psychopathy), while dimensions with more ambiguous or overlapping semantics require greater methodological caution.

Taken together, these results suggest that PFA should not be conceptualized as a standalone validation method, particularly in cross-linguistic contexts. Instead, its primary value lies in its capacity to support early-stage psychometric decision-making by identifying plausible factor structures, highlighting semantically ambiguous items, and informing subsequent empirical data collection. When used in conjunction with traditional validation approaches, PFA may substantially reduce development costs and guide hypothesis formulation, while preserving the central role of human response data in establishing psychometric validity. From an applied perspective, the present findings should be interpreted in light of the exploratory nature of the proposed approach. Rather than supporting confirmatory use in clinical decision-making, embedding-based PFA may offer practical advantages at early stages of instrument development. These include rapid prototyping of new scales, preliminary inspection of dimensional structure prior to data collection, and low-cost screening of potential translation issues or semantic inconsistencies before undertaking full empirical validation. In this sense, PFA is best conceptualized as a complementary analytic tool that can inform and streamline subsequent psychometric testing, rather than as a substitute for established validation procedures.

Finally, it should be noted that, while Pseudo-Factor Analysis (PFA) represents a logical extension of Exploratory Factor Analysis (EFA) using PLM-based item embeddings, it is merely one of many potential applications within this emerging paradigm. The methodology proposed here is grounded in the fundamental assumption that item-level embeddings can effectively serve as proxies for human response patterns. Our findings demonstrate that, within specific linguistic and thematic contexts, this assumption holds true, enabling the use of PFA as a viable psychometric tool. The extent to which embedding-based PFA may generalize to all types of psychological measures, however, remains an open question. Measures such as cognitive or ability tests often rely less on semantic similarity between item wordings and more on task demands, problem-solving processes, or performance-based scoring, which may limit the relevance of embedding-derived representations. Conversely, attitude or belief scales, which share a stronger dependence on explicit verbal content, may represent a more plausible extension of the present approach.

Accordingly, the applicability of PFA is likely to be domain-dependent, with greater relevance for instruments whose latent structure is closely tied to linguistic formulation and conceptual overlap among items.

Nevertheless, embracing the broader premise—that item embeddings can substitute for human-generated data—opens the door to a wide array of psychometric methodologies that extend beyond traditional EFA. Since correlation matrices derived from item embeddings are not inherently tied to EFA, they can be applied within alternative frameworks that rely on inter-item relationships. For example,

applications extend to psychological network analysis (Russell-Lasalandra et al., 2024) to utilize item embeddings to model the dynamic interactions between constructs. Similarly, Feraco & Toffalini (2025) proposed an application to Structural Equation Modeling (SEM) to integrate these embeddings and explore complex latent variable structures without the need for initial human response data. An important aspect that follows the unrequired human responses is that there is no sample size. A sample size is necessary for testing the significance of model parameters (e.g., factor loadings' significance) as well as determining model-based fit tests. We do not recommend simply assuming an arbitrary sample size, because this directly impacts the fit statistics and tests. However, one can consider exploratory as well as residual-based fit measures to assess the models. For instance, Suárez-Álvarez et al. (2026), suggest using Root Mean Square Residual and the Common Part Accounted for (CAF, Lorenzo-Seva et al., 2011). While this study focused on PFA due to EFA's widespread use and foundational role in psychometrics, researchers should not view EFA as the sole avenue for scale development when working with PLM-generated embeddings. In fact, more sophisticated and flexible methodologies may offer deeper insights and greater utility, particularly when aiming to capture the complexity of psychological constructs. Future research should explore these expanded possibilities, further harnessing the power of PLMs to innovate and streamline psychometric practice.

An important caveat is that PFA should not be considered a substitute for empirical validation. Over-reliance on data-free methods risks premature adoption of psychometric tools without sufficient testing on human participants, raising both scientific and ethical concerns.

4.1. Limitations

This study advanced the understanding of PFA's strengths and limitations in psychometric validation. While the method demonstrates considerable potential, particularly for English-language psychological measures, its current limitations in cross-lingual contexts and complex construct recovery indicate that further methodological refinement is needed. These results align with recent developments in embedding-based psychometrics, such as Wulff and Mata (2024) on construct overlap, Feraco and Toffalini (2025) on SEMbeddings, and Jung and Seo (2025) on data-free item reduction, all of which similarly reported strong replication of factor structures in English scales and more variable performance in other languages. Future research should focus on enhancing PFA's robustness through cross-lingual and construct-specific fine-tuning. PFA's capacity to streamline psychometric validation positions it as a valuable tool in the evolving landscape of psychological assessment.

Several general methodological limitations should be acknowledged when interpreting the present findings. First, pretrained text embeddings inevitably reflect biases present in the corpora and objectives used during model training, which may influence semantic representations in ways that are not directly aligned with psychometric constructs. Second, the PFA approach implicitly assumes that semantic similarity between item texts approximates latent covariance structures observed in empirical data—an assumption that may not hold uniformly across constructs, domains, or item formulations.

Third, in cross-language applications, translation quality and cultural adaptation may substantially alter item meaning, thereby affecting embedding geometry independently of the underlying construct. These factors introduce sources of variability that are difficult to disentangle within an embedding-based framework and further support the interpretation of PFA as an exploratory, rather than a confirmatory, analytic approach.

A further conceptual limitation concerns the potential mismatch between human and model-based representations of meaning. While human responses to questionnaire items are shaped by subjective experience, introspection, and contextual grounding (i.e., descriptive

semantics), pretrained language models encode statistically regularized patterns of language use that may reflect normative or prescriptive aspects of meaning. As a result, high semantic similarity in embedding space does not necessarily imply equivalence in experiential interpretation or response behavior.

This distinction may help explain instances in which embedding-based PFA captures formal semantic coherence without fully reproducing empirical covariance structures, particularly for constructs that rely on affective nuance, situational context, or implicit self-referential processing.

A key limitation of the present cross-language comparison is also the difficulty in disentangling language effects from translation quality and cultural construct validity. Differences in embedding-based PFA performance may reflect linguistic properties of the target language, translation-induced semantic shifts, or mismatches in how constructs are culturally instantiated and expressed. Without independent indices of translation equivalence or cross-cultural measurement invariance, these sources of variance cannot be fully separated.

Consequently, the present findings should be interpreted as indicative rather than diagnostic of language-specific effects, and future work would benefit from systematically incorporating translation quality assessments and culturally grounded validation frameworks.

Finally, further research is necessary to dissect model performance for embedding-based psychometrics, including identifying which model performs best (e.g., fine-tuned on psychological data) and in scenarios beyond clinical application. Future studies could explore the use of Procrustes rotation and residual analyses alongside Tucker's congruence (Schönemann, 1966; McCrae et al., 1996), an approach frequently adopted in psychometrics to evaluate the replicability of factor structures, to obtain a more fine-grained description of discrepancies between empirical and embedding-derived loading matrices.

5. Conclusions

This study broadens the application of PFA by evaluating both English and Italian versions of well-validated psychological scales in health and clinical domains. Our findings indicate that PFA is highly effective in recovering factorial structures for English instruments—particularly for constructs such as Depression and Psychopathy. Performance was less consistent for translated scales, with Anxiety and Stress dimensions proving more difficult to recover. Importantly, the high congruence observed for the MSP, originally developed in French and validated in Italian, suggests that PFA can successfully handle non-English original instruments, highlighting the influence of language origin in shaping outcomes. Overall, the results highlight both the opportunities offered by PLM-based psychometrics and the challenges that remain to be addressed.

Recent advances in embedding-based psychometrics further support these conclusions. Guenole et al. (2025) demonstrated the viability of PFA in replicating factor structures from embeddings; Wulff and Mata (2024) showed that semantic similarity predicts empirical overlap among constructs; Feraco and Toffalini (2025) introduced SEMbeddings to anticipate SEM fit; Jung and Seo (2025) leveraged T5 for data-free item reduction; and Russell-Lasalandra et al. (2024) proposed AI-GENIE for automated item development. Collectively, these studies converge with our results in underscoring the robustness of embeddings for English instruments, while emphasizing the critical need for further research in cross-linguistic contexts.

Looking ahead, future research should prioritize cross-linguistic fine-tuning of PLMs, the development of multilingual psychology-specific corpora, and hybrid approaches that integrate data-free methods with small empirical samples. These advances will both strengthen the methodological foundations of PFA and expand its applicability across diverse linguistic and cultural contexts. In doing so, PFA has the potential to become a robust and reliable tool for accelerating psychometric innovation while maintaining scientific rigor.

CRedit authorship contribution statement

Simone Varrasi: Writing – original draft, Investigation, Conceptualization. **Giuseppe Alessio Platania:** Writing – original draft, Investigation, Conceptualization. **Sabrina Castellano:** Writing – review & editing. **Francesca Ferraioli:** Writing – review & editing. **Simona Massimino:** Writing – review & editing. **Carmelo Mario Vicario:** Writing – review & editing. **Santo Di Nuovo:** Writing – review & editing. **Epifanio Damiano D'Urso:** Writing – original draft, Software, Formal analysis, Conceptualization.

Consent to participate

Is not applicable.

Ethics approval

The data used in this study are artificially generated; therefore no ethical approval was required.

Consent for publication

Not applicable.

Funding

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

the authors acknowledge support from the University of Messina through the APC initiative.

Data availability

The data and materials for all experiments are available at <https://osf.io/p2f5e/> and none of the experiments was preregistered.

References

- Agrafiotis, D.K., Yang, E., Littman, G.S., Byttebier, G., Dipietro, L., DiBernardo, A., Chavez, J.C., Rykman, A., McArthur, K., Hajjar, K., Lees, K.R., Volpe, B.T., Krams, M., Krebs, H.I., 2021. Accurate prediction of clinical stroke scales and improved biomarkers of motor impairment from robotic measurements. *PLoS One* 16 (1 January). <https://doi.org/10.1371/journal.pone.0245874>.
- Arnulf, J.K., Olsson, U.H., Nimon, K., 2024. Measuring the menu, not the food: “psychometric” data may instead measure “lingometrics” (and miss its greatest potential). *Front. Psychol.* 15. <https://doi.org/10.3389/fpsyg.2024.1308098>.
- Babcock, B., Albano, A.D., 2012. Rasch scale stability in the presence of item parameter and trait drift. *Appl. Psychol. Meas.* 36 (7). <https://doi.org/10.1177/0146621612455090>.
- Bottesi, G., Ghisi, M., Altoè, G., Conforti, E., Melli, G., Sica, C., 2015. The Italian version of the depression anxiety stress Scales-21: factor structure and psychometric properties on community and clinical samples. *Compr. Psychiatry* 60, 170–181. <https://doi.org/10.1016/j.comppsych.2015.04.005>.
- Costa Jr., P.T., McCrae, R.R., 1992. Revised NEO Personality Inventory (NEO PI-R) and NEO Five-Factor Inventory (NEO-FFI) Professional Manual. Psychological Assessment Resources.
- Di Nuovo, S., Rispoli, L., Genta, E., 2000. *Misurare lo Stress: Il Test M.S.P. E Altri Strumenti per Una Valutazione Integrata*, first ed. Franco Angeli.
- Feraco, T., Toffalini, E., 2025. SEMbeddings: how to evaluate model misfit before data collection using large-language models. *Front. Psychol.* 15. <https://doi.org/10.3389/fpsyg.2024.1433339>.

- Fillion, L., Tessier, R., Tawadros, É., Mouton, C., 1989. Stress et immunité: Étude de validité d'une mesure de stress psychologique (MSP). *Can. Psychol./Psychol. Canad.* 30 (1), 30–38. <https://doi.org/10.1037/h0079852>.
- Furr, R.M., Bacharach, V.R., 2019. Psychometrics: an introduction. *Making Sense of Statistics*.
- Gomez, R., Stavropoulos, V., Griffiths, M.D., 2020. Confirmatory factor analysis and exploratory structural equation modelling of the factor structure of the depression anxiety and stress Scales-21. *PLoS One* 15 (6). <https://doi.org/10.1371/journal.pone.0233998>.
- Guenole, N., D'Urso, E.D., Samo, A., Sun, T., Haslbeck, J.M.B., 2025. Enhancing scale development: pseudo factor analysis of language embedding similarity matrices. <https://doi.org/10.31234/osf.io/vf3se.v2>.
- Harris, Z.S., 1954. Distributional structure. *Word* 10 (2–3), 146–162. <https://doi.org/10.1080/00437956.1954.11659520>.
- Hassani, H., Silva, E.S., 2023. The role of ChatGPT in data science: how AI-Assisted conversational interfaces are revolutionizing the field. *Big Data and Cognitive Computing* 7 (2). <https://doi.org/10.3390/bdcc7020062>.
- Havaldar, S., Rai, S., Singhal, B., Liu, L., Guntuku, S.C., Ungar, L., 2023. Multilingual language models are not multicultural: a case study in emotion. In: *Proceedings of the 13th Workshop on Computational Approaches to Subjectivity, Sentiment, and Social Media Analysis (WASSA 2023)*. Association for Computational Linguistics, pp. 202–214. <https://doi.org/10.18653/v1/2023.wassa-1.19>.
- Hernandez, I., Nie, W., 2023. The AI-IP: minimizing the guesswork of personality scale item development through artificial intelligence. *Pers. Psychol.* 76 (4). <https://doi.org/10.1111/peps.12543>.
- Iannattone, S., Mignemi, G., Pivetta, E., Gatta, M., Sica, C., Cardi, V., Canale, N., Spoto, A., Bottesi, G., 2024. Are anxiety, depression, and stress distinguishable in Italian adolescents? An examination through the depression anxiety stress Scales-21. *PLoS One* 19 (2 February). <https://doi.org/10.1371/journal.pone.0299229>.
- Innes, J.M., Morrison, B.W., 2021. Artificial intelligence and psychology. In: *The Routledge Social Science Handbook of AI*. Routledge, pp. 30–57. <https://doi.org/10.4324/9780429198533-4>.
- Jonason, P.K., Webster, G.D., 2010. The dirty dozen: a concise measure of the dark triad. *Psychol. Assess.* 22 (2). <https://doi.org/10.1037/a0019265>.
- Jung, S.-J., Seo, J.-W., 2025. A transformer-based embedding approach to developing short-form psychological measures. *Front. Psychol.* 16, 1640864. <https://doi.org/10.3389/fpsyg.2025.1640864>.
- Kiers, H.A., 1994. Simplimax: oblique rotation to an optimal target with simple structure. *Psychometrika* 59 (4), 567–579.
- Kim, S., Kim, E., 2020. The use of virtual reality in psychiatry: a review. *Journal of the Korean Academy of Child and Adolescent Psychiatry* 31 (1). <https://doi.org/10.5765/jkacap.190037>.
- Kline, P., 2015. A handbook of test construction (psychology revivals) : introduction to psychometric design. In: *A Handbook of Test Construction (Psychology Revivals)*. Landauer, T.K., Dumais, S.T., 1997. A solution to Plato's problem: the latent semantic analysis theory of acquisition, induction, and representation of knowledge. *Psychol. Rev.* 104 (2), 211–240. <https://doi.org/10.1037/0033-295X.104.2.211>.
- Landers, R.N., Behrend, T.S., 2022. Auditing the AI auditors: a framework for evaluating fairness and bias in high stakes AI predictive models. *Am. Psychol.* 78 (1). <https://doi.org/10.1037/amp0000972>.
- Laverghetta, A., Licato, J., 2023. Generating better items for cognitive assessments using large language models. *Proceedings of the Annual Meeting of the Association for Computational Linguistics*. <https://doi.org/10.18653/v1/2023.bea-1.34>.
- Laverghetta, A., Nighojkar, A., Mirzakhlov, J., Licato, J., 2022. Predicting human psychometric properties using computational language models. In: *Springer Proceedings in Mathematics and Statistics*, 393. https://doi.org/10.1007/978-3-031-04572-1_12.
- Lê, S., Josse, J., Husson, F., 2008. FactoMineR: an R package for multivariate analysis. *J. Stat. Software* 25 (1). <https://doi.org/10.18637/jss.v025.i01>.
- Lemyre, L., Lalande-Markon, M.P., 2009. Psychological stress measure (PSM-9): integration of an evidence-based approach to assessment, monitoring, and evaluation of stress in physical therapy practice. *Physiother. Theory Pract.* 25 (Issues 5–6). <https://doi.org/10.1080/09593980902886321>.
- Lemyre, L., Tessier, R., 1988. Mesure de Stress Psychologique (MSP): Se sentir stressé-e. *Can. J. Behav. Sci./Rev. Can. Sci. Comport.* 20 (3). <https://doi.org/10.1037/h0079945>.
- Levisen, C., 2022. Postcolonial semantics. *Scandinavian Studies in Language* 13 (1), 67–77. <https://doi.org/10.7146/sss.v13i1.135072>.
- Lopes, S.L., Ferreira, A.I., Prada, R., Schwarzer, R., 2023. Social robots as health promoting agents: an application of the health action process approach to human-robot interaction at the workplace. *Int. J. Hum. Comput. Stud.* 180, 103124. <https://doi.org/10.1016/j.ijhcs.2023.103124>.
- Lorenzo-Seva, U., ten Berge, J.M.F., 2006. Tucker's congruence coefficient as a meaningful index of factor similarity. *Methodology* 2 (2). <https://doi.org/10.1027/1614-2241.2.2.57>.
- Lorenzo-Seva, U., Timmerman, M.E., Kiers, H.A., 2011. The hull method for selecting the number of common factors. *Multivariate Behav. Res.* 46 (2), 340–364. <https://doi.org/10.1080/00273171.2011.564527>.
- McCrae, R.R., Zonderman, A.B., Costa Jr., P.T., Bond, M.H., Paunonen, S.V., 1996. Evaluating replicability of factors in the revised NEO personality inventory: confirmatory factor analysis versus procrustes rotation. *J. Pers. Soc. Psychol.* 70 (3), 552–566. <https://doi.org/10.1037/0022-3514.70.3.552>.
- Moss, A.J., Rosenzweig, C., Robinson, J., Jaffe, S.N., Litman, L., 2023. Is it ethical to use mechanical turk for behavioral research? Relevant data from a representative survey of MTurk participants and wages. *Behav. Res. Methods* 55 (8). <https://doi.org/10.3758/s13428-022-02005-0>.
- Moustafa, M.A.M., Mohamed, W.M.A., Lau, A.C.C., Chatanga, E., Qiu, Y., Hayashi, N., Naguib, D., Sato, K., Takano, A., Matsuno, K., Nonaka, N., Taylor, D.M., Kawabata, H., Nakao, R., 2020. R A language and environment for statistical computing. *R Foundation for Statistical Computing*, 20.
- Ni, J., Abrego, G.H., Constant, N., Ma, J., Hall, K., Cer, D., Yang, Y., 2022. May. Sentence5: Scalable sentence encoders from pre-trained text-to-text models. In *Findings of the association for computational linguistics. ACL 2022*, 1864–1874.
- Orrù, G., Monaro, M., Conversano, C., Gemignani, A., Sartori, G., 2020. Machine learning in psychometrics and psychological research. *Front. Psychol.* 10. <https://doi.org/10.3389/fpsyg.2019.02970>.
- Petrich, L., Jin, J., Dehghan, M., Jagersand, M., 2022. A quantitative analysis of activities of daily living: insights into improving functional Independence with assistive robotics. *Proceedings - IEEE International Conference on Robotics and Automation*. <https://doi.org/10.1109/ICRA46639.2022.9811960>.
- Reimers, N., Gurevych, I., 2019. Sentence-BERT: Sentence Embeddings Using Siamese BERT-networks.
- Revelle, M.W., 2017. *Psych: Procedures for Personality and Psychological Research (R Package)*. October.
- Russell-Lasalandra, L.L., Christensen, A.P., Golino, H., 2024. Generative psychometrics via AI-GENIE: automatic item generation and validation via network-integrated evaluation. <https://doi.org/10.31234/osf.io/fgbj4>.
- Schimmenti, A., Jonason, P.K., Passanisi, A., La Marca, L., Di Dio, N., Gervasi, A.M., 2019. Exploring the dark side of personality: emotional awareness, empathy, and the dark triad traits in an Italian sample. *Curr. Psychol.* 38 (1). <https://doi.org/10.1007/s12144-017-9588-6>.
- Schönemann, P.H., 1966. A generalized solution of the orthogonal procrustes problem. *Psychometrika* 31 (1), 1–10. <https://doi.org/10.1007/BF02289451>.
- Suárez-Álvarez, J., He, Q., Guenole, N., D'Urso, D., 2026. Using artificial intelligence in test construction: a practical guide. *Psicothema* 38 (1), 1–12. <https://doi.org/10.70478/psicothema.2026.38.01>.
- Tausczik, Y.R., Pennebaker, J.W., 2010. The psychological meaning of words: LIWC and computerized text analysis methods. *J. Lang. Soc. Psychol.* 29 (1). <https://doi.org/10.1177/0261927X09351676>.
- Toth, L., Hoffmann, I., Gosztolya, G., Vincze, V., Szatlóczki, G., Bánréti, Z., Pákási, M., Kálmán, J., Hoffmann, M., 2018. A speech recognition-based solution for the automatic detection of mild cognitive impairment from spontaneous speech. *Curr. Alzheimer Res.* 15 (2), 130–138. <https://doi.org/10.2174/1567205014666171121114930>.
- Tourangeau, R., Rips, L.J., Rasinski, K., 2000. *The Psychology of Survey Response*. Cambridge University Press. <https://doi.org/10.1017/CBO9780511819322>.
- Tucker, L.R., 1951. A method for synthesis of factor analysis studies. In: *Personnel Research Section Report*, 984.
- Van Rossum, G., Drake, F.L., 2009. *Python 3 Reference Manual*; CreateSpace. Scotts Valley, CA.
- Varrasi, S., Lucas, A., Soranzo, A., McNamara, J., Di Nuovo, A., 2019. IBM cloud services enhance automatic cognitive assessment via human-robot interaction. In: *Carbone, G., Ceccarelli, M., Pislá, D. (Eds.), New Trends in Medical and Service Robotics, Mechanisms and Machine Science*, 65. Springer, Cham. https://doi.org/10.1007/978-3-030-00329-6_20.
- Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A.N., Kaiser, Ł., Polosukhin, I., 2017. Attention is all you need. *Adv. Neural Inf. Process. Syst.* 2017–December.
- Wang, L., Yang, N., Huang, X., Yang, L., Majumder, R., Wei, F., 2024. Improving Text Embeddings with Large Language Models. In: *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (ACL 2024)*. Association for Computational Linguistics, pp. 11897–11916. <https://doi.org/10.18653/v1/2024.acl-long.642>.
- Wickham, H., Bryan, J., RStudio, Kalinski, M., Valery, K., Leittenne, C., Colbert, B., Hoerl, D., Miller, E., 2019. Readxl: read excel files. R package version 1.3.1. <https://CRAN.R-project.org/package=readxl>.
- Wulff, D.U., Mata, R., 2024. Automated jingle-jangle Detection: Using Embeddings to Tackle Taxonomic Incommensurability. *Preprint*.